

Complex Numbers: Absolute Value & Trigonometric Form

Precalculus Worksheet · Grade 11–12

Name: _____

Date: _____

Learning Objectives

- Graph complex numbers on the complex plane using real and imaginary axes
- Calculate the absolute value (modulus) of a complex number using the distance formula
- Convert complex numbers from rectangular form to trigonometric (polar) form

Problems

1. Identify the real part and imaginary part of the complex number below.

$$7 - 4i$$

2. Plot the complex number below by identifying the coordinates on the complex plane. State the ordered pair (a, b) that represents its location.

$$3 + 5i$$

3. Find the absolute value of the complex number below.

$$4 + 3i$$

4. Find the absolute value of the complex number below.

$$5 - 12i$$

5. Find the absolute value of the complex number below. Leave your answer in simplified radical form.

$$-3 + 7i$$



6. Determine which of the two complex numbers below is closer to the origin by comparing their absolute values.

$$z_1 = -4 + 3i \quad \text{and} \quad z_2 = 1 - 6i$$

7. Convert the complex number below to trigonometric (polar) form. Round the angle to the nearest degree.

$$1 + i$$

8. Convert the complex number below to trigonometric (polar) form. Round the angle to the nearest tenth of a degree.

$$3 - 4i$$

9. Convert the complex number below to trigonometric (polar) form. Round the angle to the nearest tenth of a degree.

$$-2 - 2\sqrt{3}i$$

10. A complex number z has absolute value 6 and its angle θ is 150 degrees. Write z in rectangular form ($a + bi$). Leave exact values using radicals where necessary.

$$z = 6(\cos 150^\circ + i\sin 150^\circ)$$



Complex Numbers: Absolute Value & Trigonometric Form — Answer Key

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Answer Key

1. Answer: Real part: 7, Imaginary part: -4

- A complex number is written as $a + bi$, where a is the real part and b is the imaginary part.
- Here $a = 7$ and $b = -4$.

2. Answer: (3, 5)

- The real part $a = 3$ is plotted on the horizontal (real) axis.
- The imaginary part $b = 5$ is plotted on the vertical (imaginary) axis.
- The point is at (3, 5) on the complex plane.

3. Answer: 5

- Use the formula $|z| = \sqrt{a^2 + b^2}$.
- $|z| = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$.

4. Answer: 13

- Use the formula $|z| = \sqrt{a^2 + b^2}$.
- $|z| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

5. Answer: $\sqrt{58}$

- Use the formula $|z| = \sqrt{a^2 + b^2}$.
- $|z| = \sqrt{(-3)^2 + 7^2} = \sqrt{9 + 49} = \sqrt{58}$.
- $\sqrt{58}$ cannot be simplified further.

6. Answer: z_1 is closer ($|z_1| = 5$, $|z_2| = \sqrt{37} \approx 6.08$)

- $|z_1| = \sqrt{(-4)^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$.
- $|z_2| = \sqrt{1^2 + (-6)^2} = \sqrt{1 + 36} = \sqrt{37} \approx 6.08$.
- Since $5 < 6.08$, $z_1 = -4 + 3i$ is closer to the origin.

7. Answer: $z = \sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$

- Find r : $r = \sqrt{1^2 + 1^2} = \sqrt{2}$.
- Find θ : $\theta = \arctan\left(\frac{b}{a}\right) = \arctan\left(\frac{1}{1}\right) = 45^\circ$.
- Since the complex number is in Quadrant I, $\theta = 45^\circ$.
- Write in trig form: $z = \sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$.

8. Answer: $z = 5(\cos 306.9^\circ + i \sin 306.9^\circ)$

- Find r : $r = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
- Find the reference angle: $\alpha = \arctan\left(\frac{4}{3}\right) \approx 53.1^\circ$.
- The complex number is in Quadrant IV ($a > 0$, $b < 0$), so $\theta = 360^\circ - 53.1^\circ = 306.9^\circ$.





- Write in trig form: $z = 5(\cos 306.9^\circ + i\sin 306.9^\circ)$.

9. Answer: $z = 4(\cos 240^\circ + i \sin 240^\circ)$

- Find r : $r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$.
- Find the reference angle: $\alpha = \arctan\left(\frac{2\sqrt{3}}{2}\right) = \arctan(\sqrt{3}) = 60^\circ$.
- The complex number is in Quadrant III ($a < 0$, $b < 0$), so $\theta = 180^\circ + 60^\circ = 240^\circ$.
- Write in trig form: $z = 4(\cos 240^\circ + i\sin 240^\circ)$.

10. Answer: $z = -3\sqrt{3} + 3i$

- Use $a = r\cos\theta$ and $b = r\sin\theta$.
- $a = 6\cos 150^\circ = 6 \cdot \left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$.
- $b = 6\sin 150^\circ = 6 \cdot \frac{1}{2} = 3$.
- Write in rectangular form: $z = -3\sqrt{3} + 3i$.

